SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I S. SHAMAH ALI FATHIMA (192573002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

1) COMPARE TYPE-1 AND TYPE-2 HYPERVISORS FOR A CLOUD DATACENTER HOSTING MISSION-CRITICAL APPLICATION.

The hypervisor is software that creates & manages virtual machines.

Feature	Type-1 Hypervisor	Type-2 Hypervisor.
Architecture	Runs directly on hardware	Runs on a host operating system.
Performance	High performance	Comparatively lower
Security	Smaller attack surface	Depends on host OS security.
Manageability	Suitable for large datacenters	Suitable for personal use and testing.

Analysis:

Type-1 Hypervisors:

* Can access hardware directly, reducing overhead and improving performace.

* They provide strong isolation between virtual machine are suitable for running critical business applications.

Type-2 hypervisors.

* They depend on a host operating system.

* This introduces additional overhead and makes their security dependant on the host OS.

Conclusion:

* Type-1 hypervisor is a better choice for a cloud datacenter because it offers better performance, security, scalability and reliability.

2) ANALYZE HOW VIRTUALIZATION SUPPORTS WORKLOAD ISOLATION & IDENTIFY TWO MAJOR ATTACK SURFACES.

Virtualization allows multiple virtual machines to run on the same physical hardware while keeping their workloads separate

How virtualization supports isolation.

* VM isolation:

Each virtual machine runs its own operating system and application.

* Resource isolationtion.

CPU, memory and storage resources are allocated seperately to prevent one workload from affecting another.

* Network isolation:

Virtual networks, VLAN's and firewalls restrict communication between different workloads.

* Access control:

Role-based access control ensures that only authorized users can access sensitive resources.

Two major attack surfaces:

* Hypervisor vulnerabilities:

Attackers may exploit hypervisor weaknesses to access other virtual machines.

* Virtual network vulnerabilities.

Incorrect network configurations may allow unauthorized communication between workloads.

Conclusion:

Strong VM isolation, regular security updates, network segmentation and access control help to protect sensitive finance, healthcare and student data.

3) ANALYZE THE VM HOST UTILIZATION SNAPSHOT AND RECOMMEND CORRECTIVE ACTIONS.

Given data:

* CPU utilization = 92%.
* Memory utilization = 88%.
* Storage latency = 35 ms
* Average VM boot time = 170 seconds.

Analysis & bottlenecks:

Parameter	Analysis	Corrective actions.
CPU - 92%.	Very high utilization; CPU resources may be insufficient.	Add CPU capacity or migrate VMs to less-loaded hosts.
Memory - 88%.	High memory usage may cause swapping	Increase RAM or reduce unnecessary VM memory allocation.

Parameter	Analysis	Corrective action.
Storage latency - 35 ms	High latency may slow disk operations.	Use faster storage.
VM boot time - 170s	slow startup may result from CPU	Optimize VM images, investigate startup services

Recommended corrective actions.

* Use load balancing to distribute VMs across multiple hosts.

* Monitor CPU, memory and storage performance continuously

* Apply resource reservations and limits to prevent resource contention.

CONCLUSION:

High CPU and memory utilization, together with storage latency, indicate possible resource contention. Optimizing resource allocation and increasing capacity where necessary should improve VM boot time and overall performance.

4) ANALYSE HOW SOFTWARE DEFINED DATACENTER ARCHITE... IMPROVES AGILITY COMPARED WITH A TRADITIONAL DATACENTER.

A Software Defined Datacenter is a datacenter in which computing, storage and networking resources are virtualized and managed through software.

Feature	Traditional Datacenter	Software Defined Datacenter
Compute	Physical server configuration	Virtual machines and automated resource allocation.
Storage	Dedicated physical storage systems	Software-defined storage pools.
Network	Hardware-based configuration.	Software-defined networking.
Management	Mostly manual	Centralized and automated.
Scalability	Requires more manual setup.	Resources can be Provisioned quickly.

Conclusion:

SDDC improves agility by enabling fastes deployment, centralized mangement, automation and flexcible resource allocation. Compared with traditional datacenters, it responds more quickly to changing business needs.

5) ANALYZE IDENTITY MISHATCHES IN A MULTI-CLOUD FEDERATION PROJECT AND PROPOSE A SECURE IDENTITY AND PRIVACY DESIGN.

Problem analysis:

* A multi-cloud federation connects services from different cloud providers so they can work together.

* Identity mismatches occur when providers use different user identifiers, authentication methods, roles or Permission structures.

Proposed secure identity and privacy design:

* Federated identity management:

Use a trusted identity provider to manage user identities across cloud platforms.

* Single Sign-On:

Allow users to access multiple authorized cloud services using one authentication process.

* Standard protocols:

Use SAML 2.0 or OpenID Connect for federated authentication with OAuth 2.0 for delegated authorization where appropriate.

CONCLUSION:

A secure multi-cloud federation should use standardized identity protocols, consistent role mapping, MFA, encryption and continuous monitoring.